DO NOW — RECALL

Lesson 127 · Lossless and Lossy Compression

Name: _____ Class: _____ Date: _____

1. What is indirect addressing in a CPU instruction?

2. What is the binary result of 1 + 1 in a full adder with carry-in 0?

3. What is O(n) search complexity associated with?

4. Which memory is typically volatile: RAM or ROM?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

The instruction points to a location holding the effective address.

2. Model answer

Sum 0, carry-out 1.

3. Model answer

Linear/sequential search in the worst case.

4. Model answer

RAM.

By the end, you can…

• Repeated patterns and predictable structures can be encoded efficiently.

• Lossless retains all detail; lossy permits irreversible approximation.

• Consider file size, compression ratio, quality and purpose.

Vocabulary — know these terms

lossless — Original data can be reconstructed exactly.

lossy — Some original information is deliberately discarded.

redundancy — Repeated or predictable data.

compression ratio — Original size divided by compressed size (by this convention).

NEW LEARNING

Read, annotate and explain

Find redundant data

Repeated patterns and predictable structures can be encoded efficiently.

Choose method

Lossless retains all detail; lossy permits irreversible approximation.

Measure impact

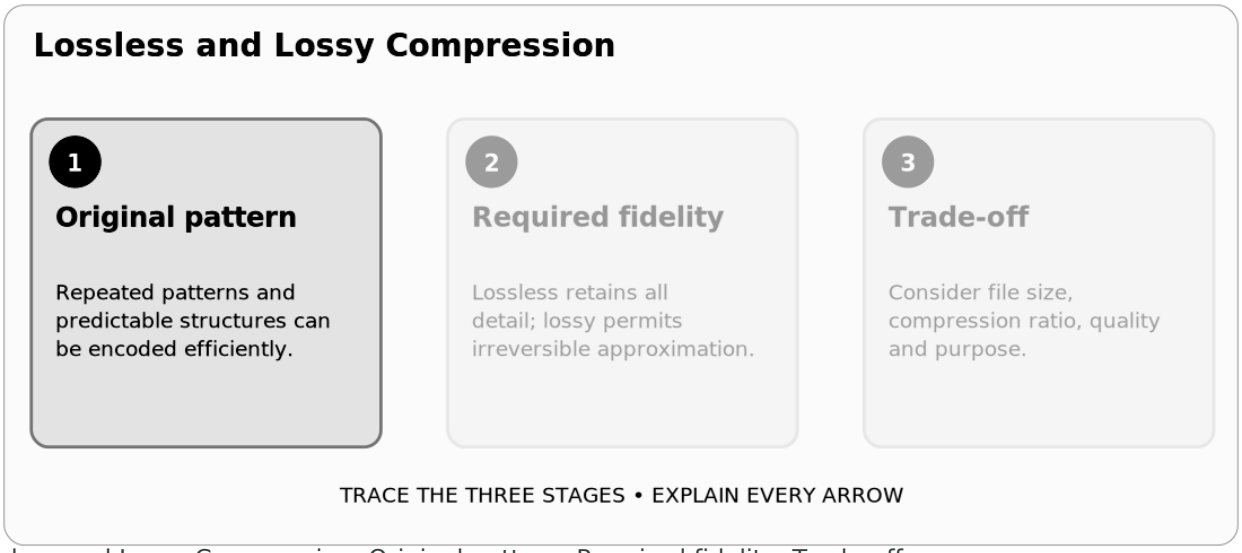
Consider file size, compression ratio, quality and purpose.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

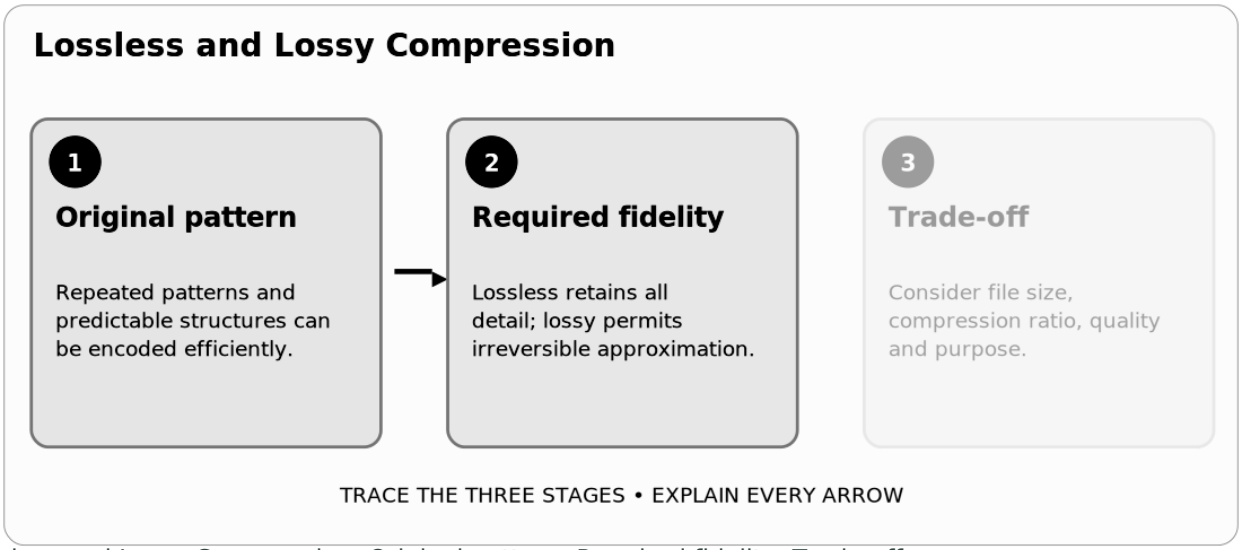
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Lossless and Lossy Compression: Original pattern; Required fidelity; Trade-off

New Learning

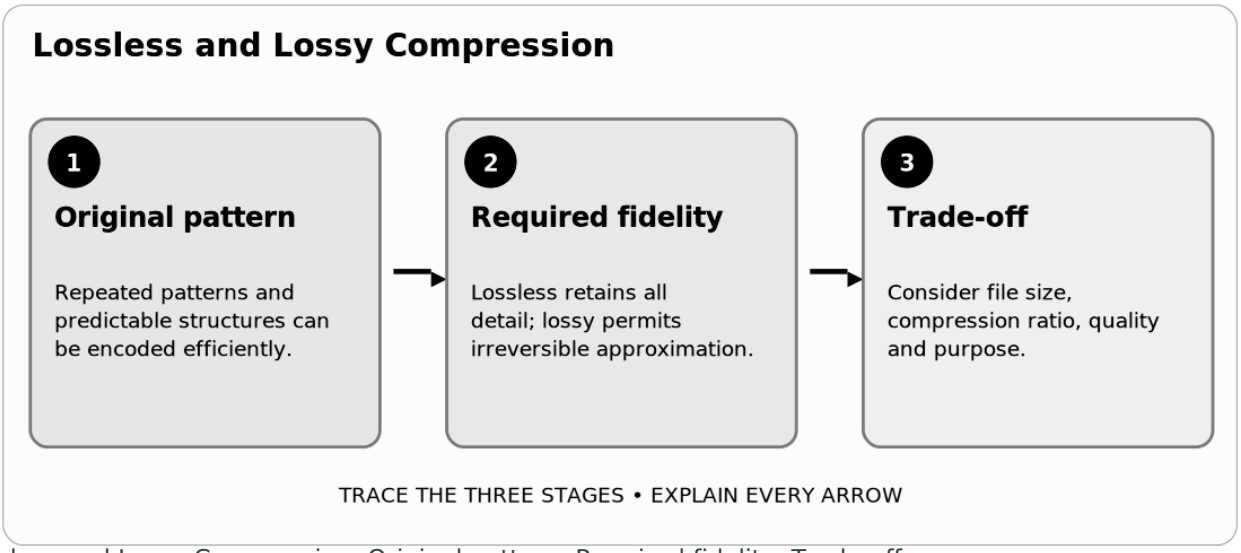


Lossless and Lossy Compression: Original pattern; Required fidelity; Trade-off

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Lossless and Lossy Compression: Original pattern; Required fidelity; Trade-off

I DO — WORKED EXAMPLE

Follow the worked method

A medical image must remain identical after decompression. Choose a technique and justify.

Medical image data

Predict the first step from the prompt.

Lossless encoding

Explain the intermediate result or decision.

Exact reconstruction

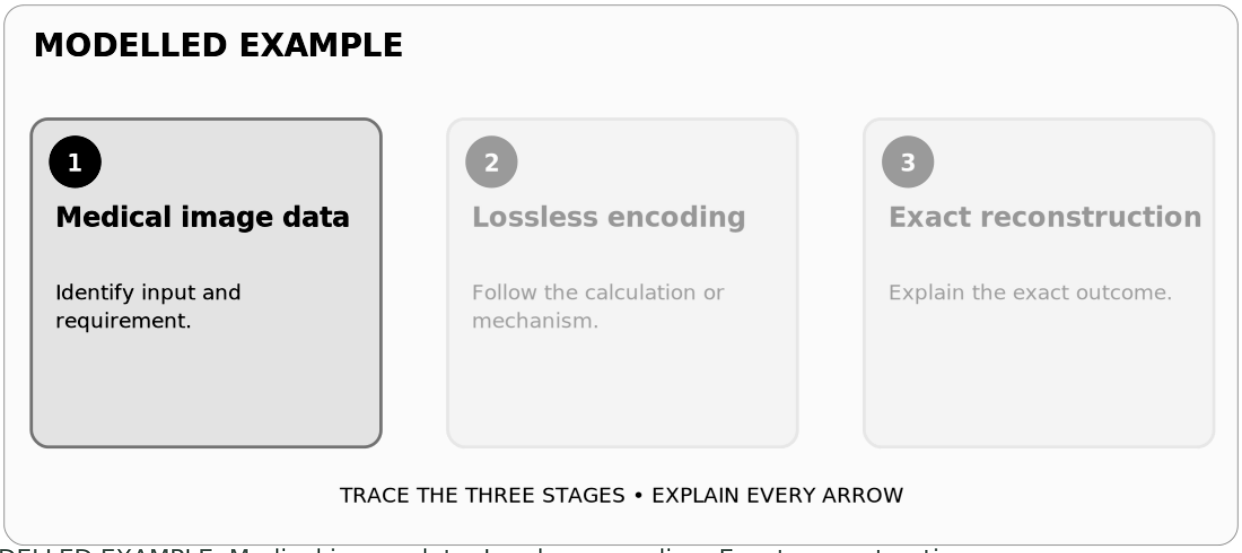
Lossless compression is required if exact original image data must be preserved. Lossy compression permanently discards information and could alter diagnostically important features.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A medical image must remain identical after decompression. Choose a technique and justify.

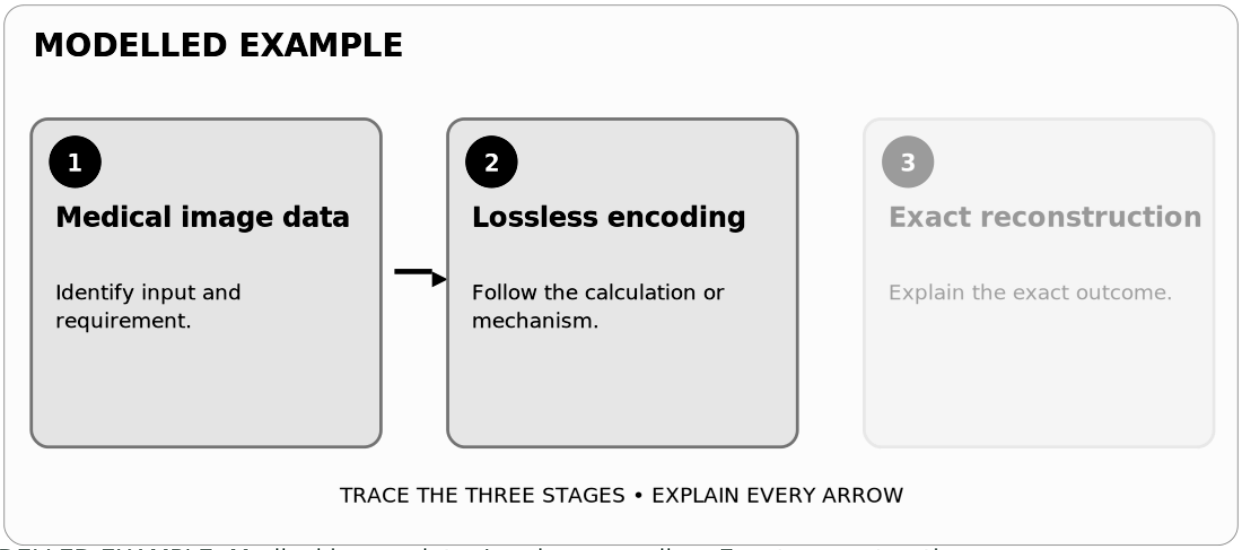
Predict the first step from the prompt.



MODELLED EXAMPLE: Medical image data; Lossless encoding; Exact reconstruction

A medical image must remain identical after decompression. Choose a technique and justify.

Predict the first step from the prompt.



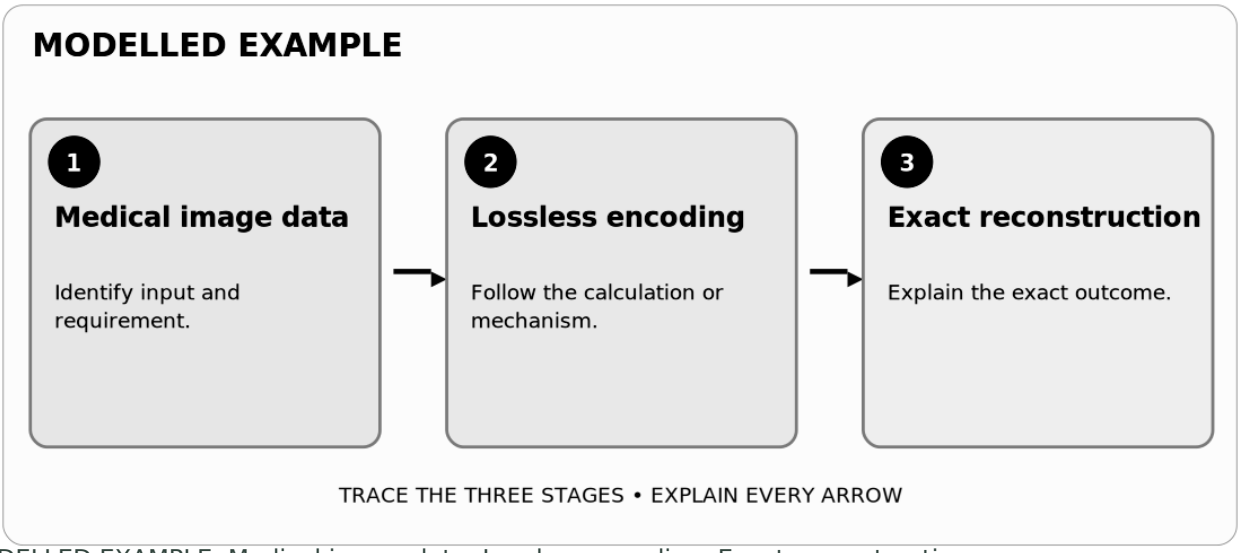
MODELLED EXAMPLE: Medical image data; Lossless encoding; Exact reconstruction

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A medical image must remain identical after decompression. Choose a technique and justify.

Predict the first step from the prompt.



MODELLED EXAMPLE: Medical image data; Lossless encoding; Exact reconstruction

WE DO — GUIDED PRACTICE

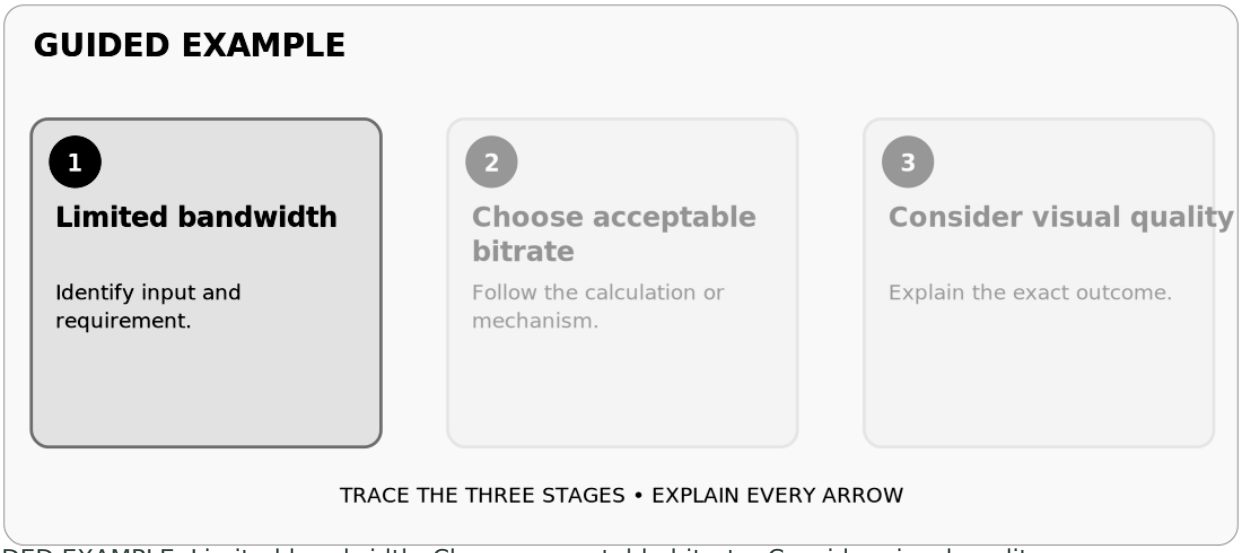
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A streaming service sends video over a limited-bandwidth connection. Is lossy compression always unacceptable?

Predict the first step from the prompt.



GUIDED EXAMPLE: Limited bandwidth; Choose acceptable bitrate; Consider visual quality

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Limited bandwidth Predict the first step from the prompt.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Define lossy and lossless compression and give one application of each. [6]

Question 2

A file is reduced from 12 MB to 3 MB. Calculate the original-to-compressed compression ratio. [4]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Evaluate a compression choice for source code, an online photograph and an MRI scan. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A lossy method always reconstructs the original bit-for-bit.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

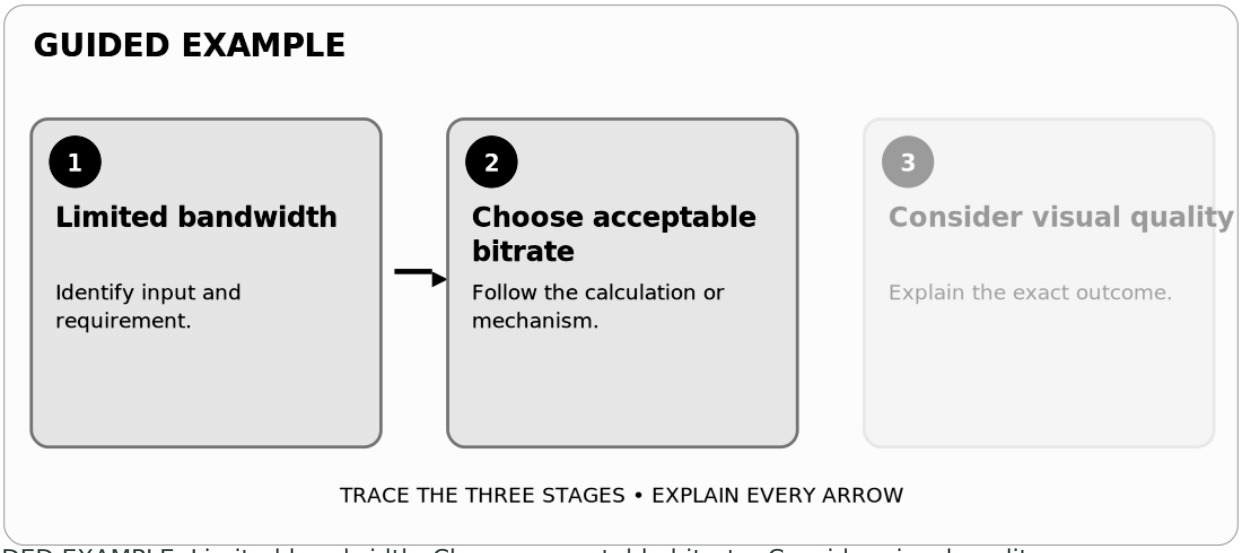
Next step: revisit the model or a relevant revision-guide section if needed.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A streaming service sends video over a limited-bandwidth connection. Is lossy compression always unacceptable?

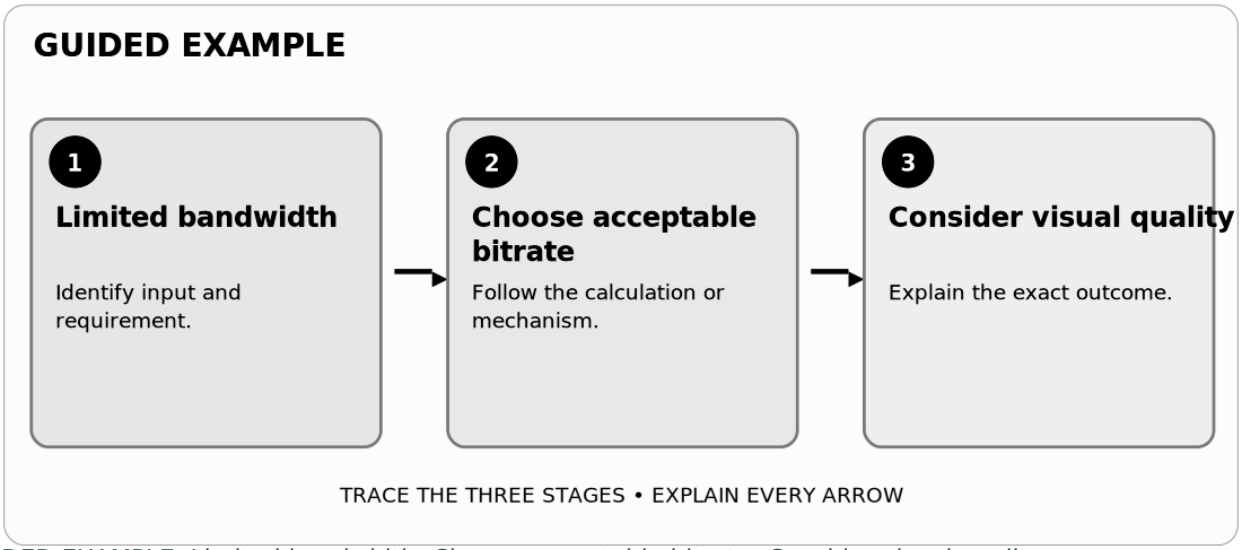
Predict the first step from the prompt.



GUIDED EXAMPLE: Limited bandwidth; Choose acceptable bitrate; Consider visual quality

A streaming service sends video over a limited-bandwidth connection. Is lossy compression always unacceptable?

Predict the first step from the prompt.



GUIDED EXAMPLE: Limited bandwidth; Choose acceptable bitrate; Consider visual quality

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Choose acceptable bitrate

Explain the intermediate result or decision.

Consider visual quality

Not necessarily. Perceptually lossy encoding can reduce bitrate enough for smooth playback, but visible artefacts may appear. Suitability depends on quality, latency and the application.

IMPROVE: Explain one correction to your own reasoning.